

SMC Trading Bot

Honest Assessment & Improvement Roadmap

What Was Fixed (v2 Upgrade)

Weak Point	Fix Applied
No backtesting	Full backtesting engine with historical replay, spread/slippage/commission, win rate, profit factor, S
Paper mode used fake data	Real market data provider: MT5 → Twelve Data API → yfinance → synthetic fallback
No entry confirmation	Zone retest + candle patterns (engulfing, pin bar, inside bar, displacement)
No trade management	Breakeven at 1R, partial close at 2R, ATR trailing stop, time-based exit
No session filtering	London, New York, overlap sessions — skip low-liquidity hours
No news filter	ForexFactory economic calendar — blocks trading during high-impact events
Basic CHoCH detection	Tracks prior trend from 3rd swing point for proper reversal detection
S/D freshness bugs	Correctly checks if price has NOT revisited the zone since departure
Settings buttons non-functional	All buttons implemented: symbols, timeframes, spread, daily loss, trades, positions, cooldown
No confirmation for dangerous actions	Close all, live mode, auto-trade ON all require explicit confirmation
DB_PATH ignored in Docker	Uses os.getenv() for Docker volume persistence

Current Strengths

Area	Rating	Notes
Risk Management	9/10	10 hard gates, position sizing, daily limits — very solid
Architecture	8/10	Clean separation, paper/live abstraction, configurable
APA Concepts	7/10	Real SMC concepts implemented correctly at intermediate level
Backtesting	7/10	Full metrics engine with realistic simulation
Entry Logic	7/10	Zone retest + candle confirmation now implemented
Trade Management	7/10	Breakeven, trailing, partial close, time exit
Data Quality	8/10	Real market data from 3 sources + MT5
Telegram UX	8/10	18 commands, full inline keyboard settings
Safety	9/10	Paper by default, confirmations, news/session filters

Remaining Limitations

These are areas that still need improvement but are not blockers:

Limitation	Impact	Priority
Scoring weights not optimized	Weights are reasonable but not backtested against real data	High — run backtests and tune
No multi-position correlation check	Bot could open correlated trades (e.g., EURUSD + GBPUSD)	Medium
No DOM/order flow data	Can't see real volume or order book depth	Low — needs broker API
Single entry strategy	Only SMC zones + confirmation; no breakout or trend-following	Medium
No walk-forward optimization	Can't auto-tune parameters on rolling windows	Low — manual tuning for now
Synthetic indices use fake data	Deriv synthetic indices have no free API	High — connect MT5 with Deriv

Recommended Workflow

1. Start in paper mode with auto-trade OFF

Use /scan and /analyze to see signals. Compare to your own analysis.

2. Run backtests on your symbols

Use /backtest EURUSD H1 180 to test 6 months of data. Check win rate, profit factor, and max drawdown. Run on multiple symbols and timeframes.

3. Tune the score threshold

Start at 40%. If too many low-quality trades, raise to 60-70%. If too few trades, lower to 30%. The backtest will tell you the optimal threshold.

4. Connect MT5 with real data

Keep paper mode but connect MT5 so analysis runs on real broker data. This makes paper testing meaningful.

5. Go live with tiny risk

0.5% risk per trade maximum. Treat it as tuition. Track every trade manually.

6. After 50-100 trades, adjust weights

See which factors correlate with winners. Adjust the scoring weights accordingly.

Backtest Commands Reference

From Telegram:

```
/backtest EURUSD H1 180 – 6 months of H1 data
```

```
/backtest XAUUSD H4 365 – 1 year of H4 data
```

```
/backtest GBPUSD M15 90 – 3 months of M15 data
```

From CLI:

```
python -m backtest.runner --symbol EURUSD --timeframe H1 --days 180 --score 50 --rr 3.0 --risk 1.0
```

Key Metrics to Watch

Metric	Good	Bad	What it means
Win Rate	>45%	<35%	Percentage of winning trades
Profit Factor	>1.5	<1.0	Gross profit divided by gross loss
Expectancy	>0	<0	Average P&L per trade
Max Drawdown	<10%	>20%	Largest peak-to-trough decline
Sharpe Ratio	>1.0	<0.5	Risk-adjusted return

Avg RR	>1.5	<1.0	Average risk-reward achieved
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